Quiz Answer Key

Category B Practice - Part B, Questions 1-20

Answers with reasons

Q1. B - protected age and public getName() are accessible within Student; private name is not.
Q2. C - Diagram 3 is invalid under single class inheritance because S extends two classes Q and R.
Q3. B - Base-to-derived constructor order: Person, Employee, Manager.
Q4. C - A derived class can directly access a protected field. It cannot access privateField, override finalMethod, or weaken public visibility.
Q5. B - Runtime dispatch invokes Bus.fare() for a Bus instance.
Q6. B - Printer's print(String) and print(int) are overloads; SpecialPrinter overrides print(String).
Q7. A - A Shape reference directly exposes members declared by Shape; Circle-specific members require a Circle reference or cast.
Q8. C - Circle.area() and Square.area() are invoked according to the runtime object.
Q9. B - Casting the Current object to Savings fails; casting it to Current succeeds.
Q10. B - Polymorphism lets different Plan subtypes be processed uniformly through a base reference.
Q11. A - printDocument() is the exposed behavior; the buffer and helper methods are hidden implementation details.
Q12. B - A Vehicle reference may reference a Car object, but abstract Vehicle cannot be instantiated.
Q13. B - A subclass missing an abstract method implementation must itself be abstract.
Q14. B - A concrete MonthlyReport without generateBody() does not compile.
Q15. A - An interface specifies operations a type promises to provide, not per-object state or full direct implementation.
Q16. B - Interface abstract methods are implicitly public abstract; fields are public static final.
Q17. C - DataProcessor must fulfill both serialize() and reset().
Q18. C - An interface is the appropriate contract for unrelated classes sharing a capability.
Q19. B - Dog IS-A Animal (generalization); Eagle CAN-DO Flyable (realization).
Q20. A - equals(), hashCode(), and toString() are common Object methods often overridden.

Compact answer string

B, C, B, C, B, B, A, C, B, B, A, B, B, B, A, B, C, C, B, A.